

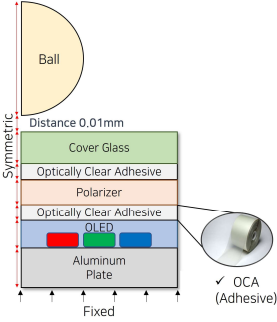
Design Optimization of Multi-layered Display Panels for Drop Impact using Physics-Constrained Graph Neural Network

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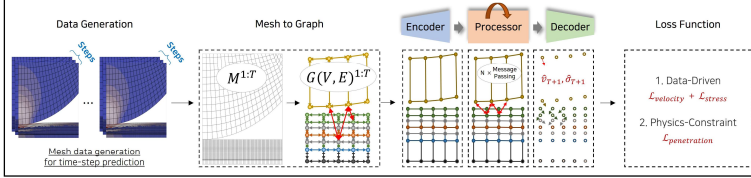
I. Introduction

- Introduction of a graph neural network-based surrogate model for predicting ball drop tests, evaluating impact resistance in multi-layered OLED panels.
- Mitigation of non-physical penetration between flexible bodies in MeshGraphNet proposed by pffaf et al. through a physics-constrained MGN, achieving over 70% reduction in position and stress errors.
- Optimization of OCA thickness using a genetic algorithm to minimize stress in the OLED panel, with a comparison of prediction consistency between baseline and proposed models.
- Emphasis on the importance of accurate stress prediction for optimal real-time results.

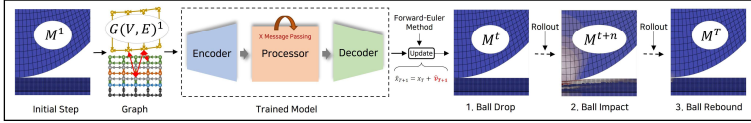


II. Physics-constrained graph neural network

<Training>

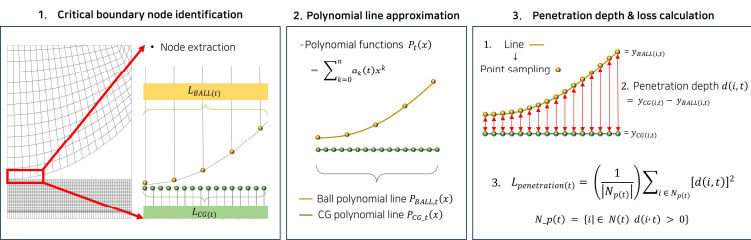


<Inference>



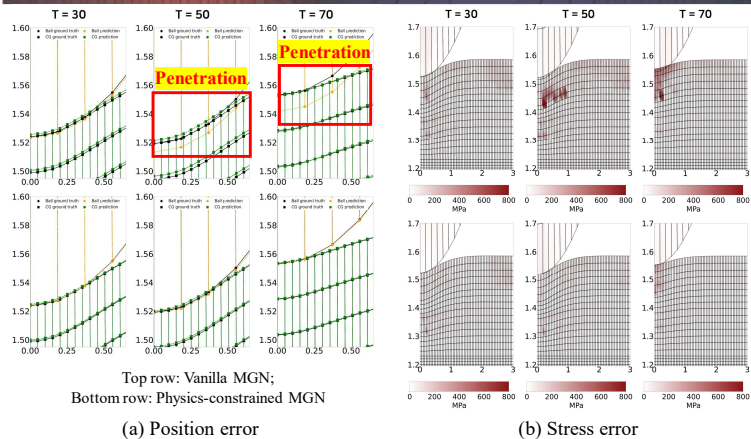
- Two-phase flowchart illustrating the training and inference processes of the physics-constrained graph neural network.

➤ Physics-constrained loss function

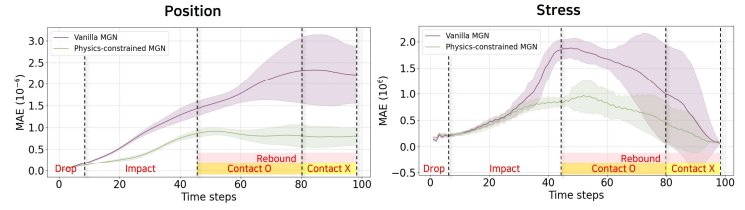


- Step 1: Extraction of critical boundary nodes for ball and cover glass
- Step 2: Polynomial line fitting for boundary approximation
- Step 3: Penetration depth calculation and corresponding loss calculation

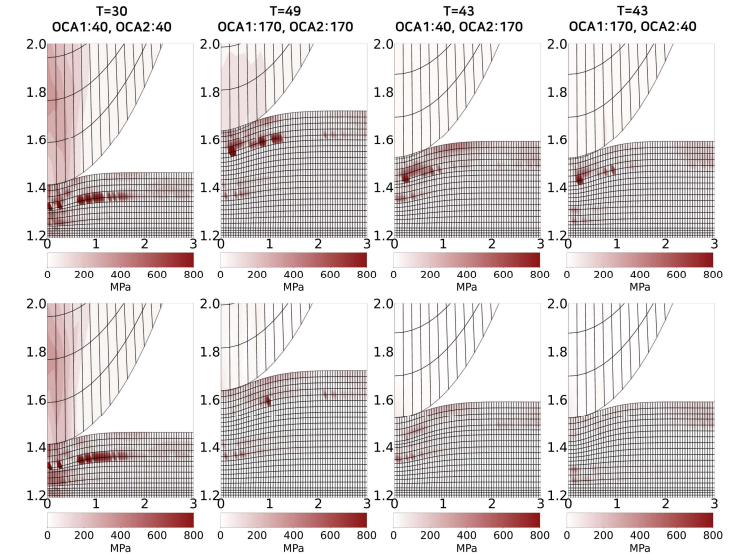
III. Results



- Penetration phenomenon in vanilla MGN, leading to delayed and inaccurate predictions.
- Stress error distribution shows the physics-constrained MGN significantly reduces errors, enhancing overall simulation accuracy.



- Higher error accumulation and fluctuations in the vanilla MGN, particularly during impact and rebound phases.
- Superior position and stress prediction accuracy in the physics-constrained MGN, with lower error accumulation and stability during all phases.

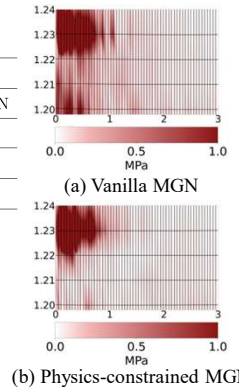


- Validation of the model's extrapolation capability beyond the training range.
- Achievement of more precise predictions with thicker OCA layers.

➤ Thickness optimization

	OCA Thickness Optimization	
	Vanilla MGN	Physics-constrained MGN
Mean stress of ground truth	1.995 MPa	1.891 MPa
Mean Stress of prediction	1.666 MPa	1.860 MPa
Error (%)	16.45	1.67

- Significant improvement in prediction consistency, with 89.8% better alignment compared to vanilla MGN
- Localized errors in the physics-constrained model, offering better accuracy compared to widespread errors in the vanilla model.



IV. Conclusion

- 800 to 1000 times speed-up over conventional numerical simulations.
- Improvement in prediction accuracy by preventing non-physical penetration with the physics-constrained MGN.
- Consistent outperformance of physics-constrained MGN with lower error rates and maintained accuracy under various conditions.
- Successful optimization of OCA thickness with better alignment to ground truth and reduced error rates.

Acknowledgement

- This work was supported by the National Research Foundation of Korea grant (2018R1A5A7025409), the Technology Innovation Program (The Alchemist Project)(RS-2024-00410810, Development of Generative Manufacturing Technologies to Turn Ideas into Reality) funded By the Ministry of Trade, Industry & Energy(MOTIE, Korea)

References

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- [2] Pfaff, T., Fortunato, M., Sanchez-Gonzalez, A., & Battaglia, P. W. (2020). Learning mesh-based simulation with graph networks. *arXiv preprint arXiv:2010.03409*.